DP26

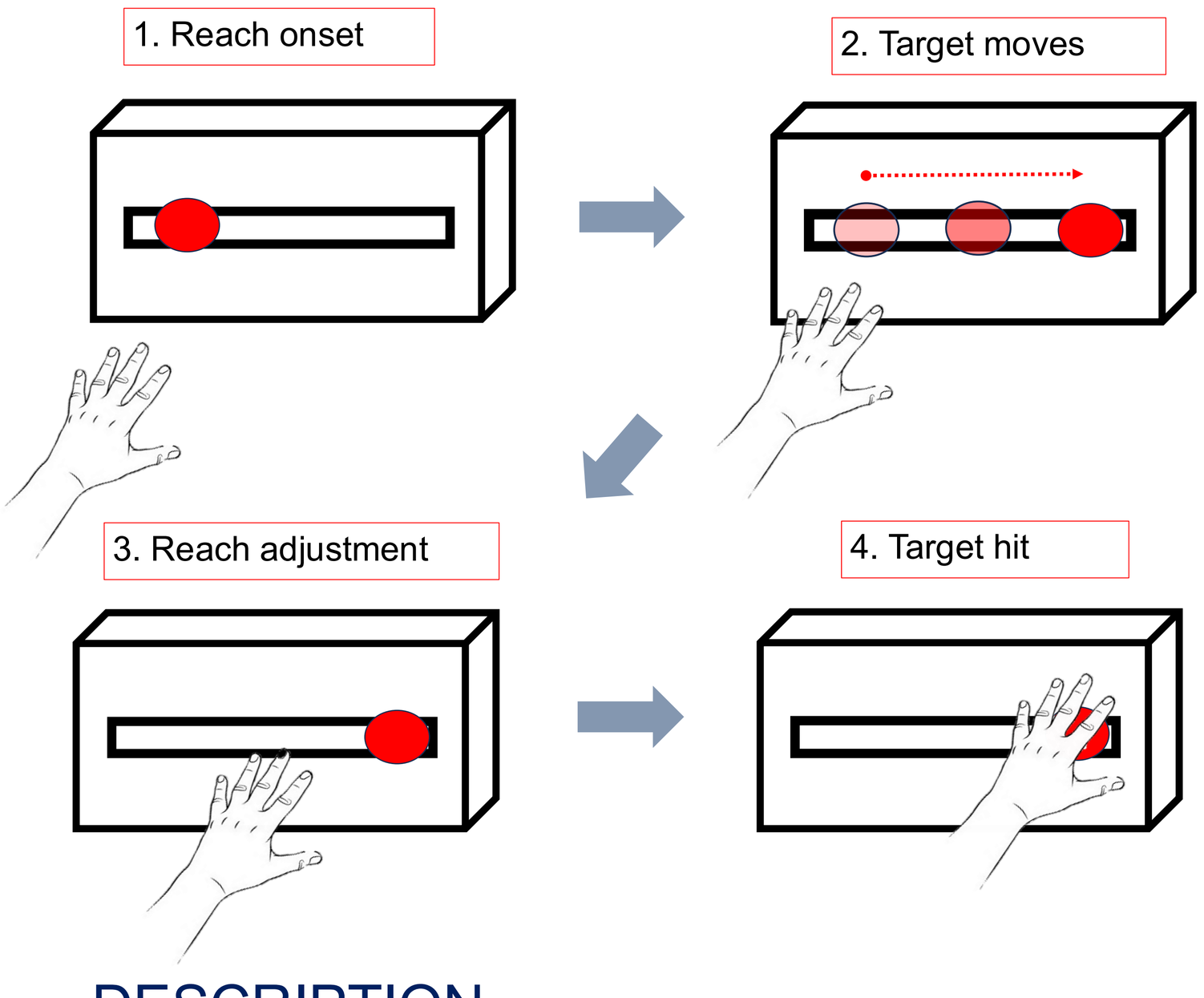
Moving Target Board for Testing Online Motor Control

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DEVELOPMENT, EXPERIENCE & MOTOR RECOVERY LAB

APPARATUS



BACKGROUND

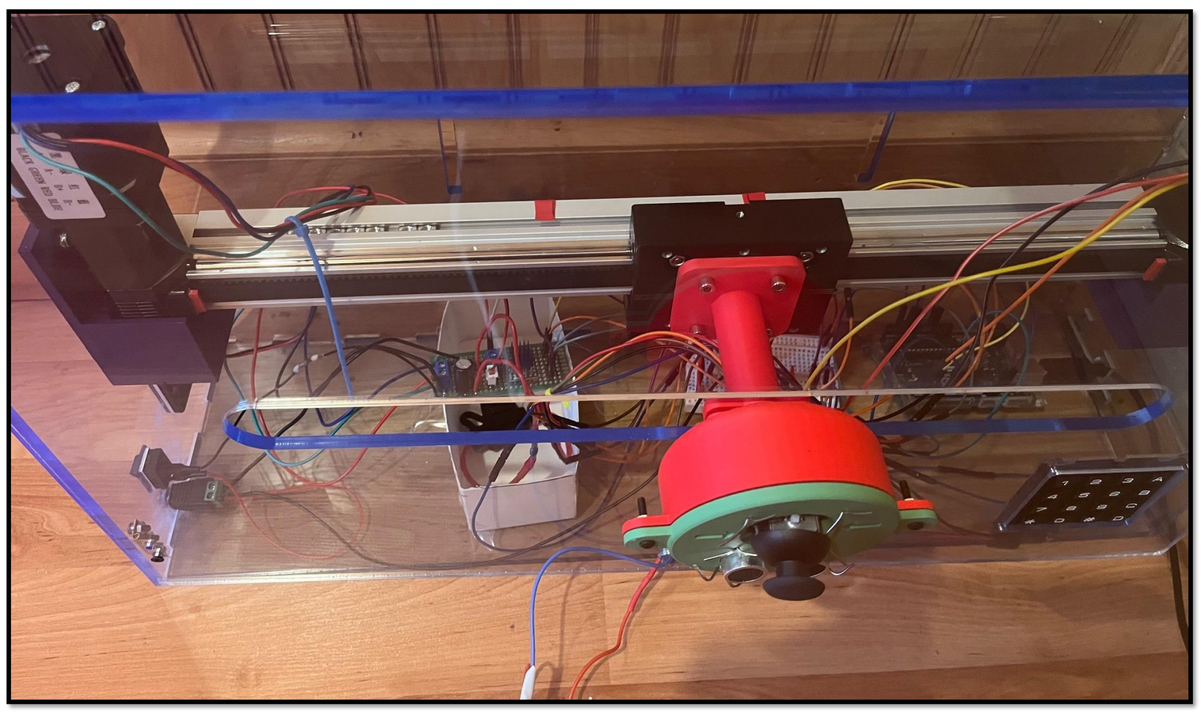
- Children with unilateral cerebral palsy demonstrate impairments in online motor control
- Current research has primarily used screen-based virtual targets
- Virtual targets lack the sensorimotor demands of physical targets, such as haptic contact and depth cues

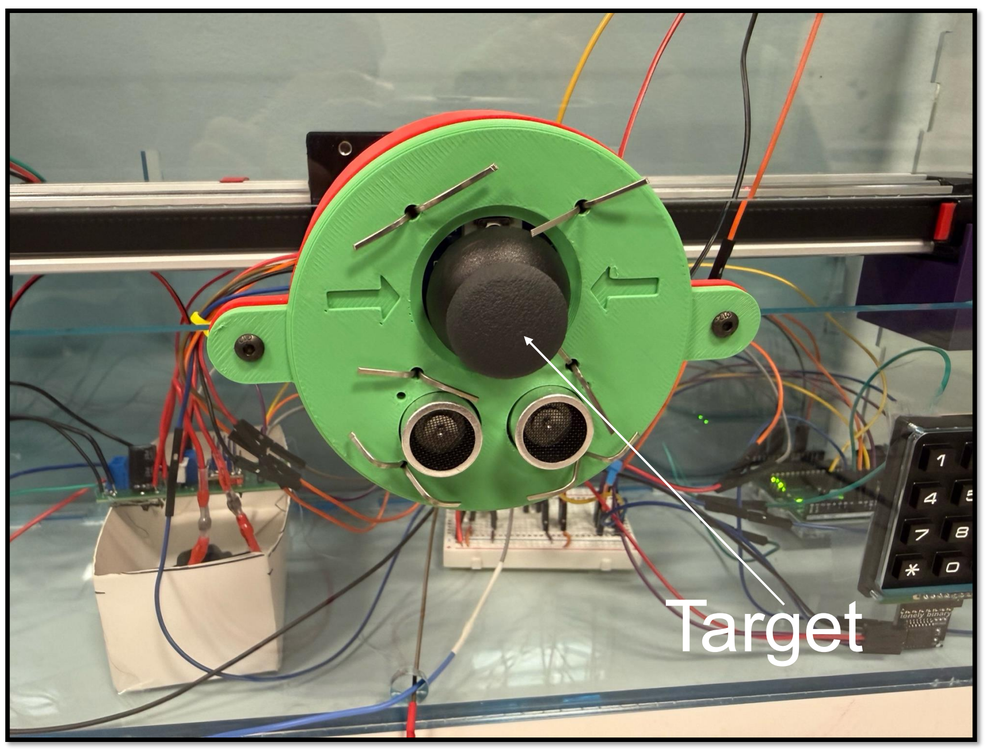
OBJECTIVE

- To demonstrate a novel physical moving-target apparatus to assess online motor control in children with UCP
- Task performance will be paired with signal obtained from IMUs and functional near-infrared spectroscopy (fNIRS)

DESCRIPTION

- Target is stationary during reach imitation on 35 cm track
- Child lifts hand off home button to reach target
- Target displaces (20 cm/sec) left or right when sensor indicates hand is 35% of the distance to target
- Measures obtained: correction latency, movement time, success rate





Video Demo

SIGNIFICANCE

- Engages sensory feedback and real 3D visual depth cues
- Enables testing whether neural activity differs between physical and virtual reaching contexts
- After validity established, can be used in motor learning and training paradigms

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